 Reg. No.

**B**

# *B.Tech. Degree II Semester Regular Examination July 2024*

## EE/EC/CS/IT 23-200-0203B DIGITAL ELECTRONICS

*(2023 Scheme)*

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the fundamental boolean functions and basic building blocks of digital systems.
CO2: Design optimal digital circuits using basic building blocks.
CO3: Analyse basic digital circuits.
CO4: Understand HDL models of simple circuits.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

### PART A
(Answer *ALL* questions)

| | | (5 × 2 = 10) | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| I. | (a) | Simplify the following Boolean expression using Boolean algebra<br>(i) $AB + \overline{AC} + A\overline{B}C(AB + C)$<br>(ii) $(X + \overline{Y} + XY)(X + \overline{Y})(\overline{X}Y)$ | 2 | L3 | 1 | 1,12 |
| | (b) | Find the minterm expansion of the following Boolean expression.<br>(i) $F(A, B, C, D) = \overline{A}(\overline{B} + D) + AC\overline{D}$<br>(ii) $F(X, Y, Z) = X(\overline{X} + Y)(\overline{X} + Y + \overline{Z})$ | 2 | L3 | 1 | 1,12 |
| | (c) | Define the figure of merit of logic gates. The following are the voltage and current parameters of a 74 series logic gate, find fan out and figure of merit.<br>Propagation delay = 9 ns,<br>Power dissipation per gate = 10 Mw,<br>$I_{OH} = 800\ \mu A$; $I_{IH} = 40\ \mu A$; $I_{OL} = 48$ mA; $I_{IL} = 1.6$ mA. | 2 | L3 | 2,3 | 1,12 |
| | (d) | Draw the timing diagram showing the loading of a 4-bit input data 0101 into a 4-bit SISO register with initial state of each flip flop in the register being logic 1. | 2 | L3 | 2,3 | 1,3,12 |
| | (e) | Draw the circuit of a 3-bit ring counter. Explain its working with timing diagram. | 2 | L2 | 2,3 | 1 |

### PART B

| | | (4 × 10 = 40) | | | | |
|---|---|---|---|---|---|---|
| II. | (a) | Determine the prime implicants of the Boolean function by using the tabulation method<br>$F(a,b,c,d) = \sum m(1, 4, 6, 7, 8, 9, 10, 11, 15)$. | 6 | L3 | 1,2,3 | 1,2,3,12 |
| | (b) | Solve using a 3 variable K map<br>(i) $AB\overline{E}\,\overline{C} + ABC\overline{D} + ABC\overline{E} + A\overline{B}C\overline{D} + \overline{A}BC$<br>(ii) $m_1 + Dm_2 + m_3 + \overline{D}m_5 + m_7 + d(m_0 + Em_1 + \overline{E}m_6)$. | 4 | L3 | 1,2 | 1,2,3,12 |
| | | **OR** | | | | |
| III. | (a) | Design a logic circuit to produce a HIGH output if and only if the input, represented by a 4-bit binary number, is greater than 12 or less than 3. Develop the truth table and draw the logic diagram. | 6 | L6 | 1,2,3 | 1,2,3,12 |
| | (b) | Calculate by 2's complement method<br>(i) $(75.75)_{10} - (126.250)_{10}$<br>(ii) $(A3.74)_{16} - (23.52)_8$. | 4 | L2 | 1,2,3 | 1,2,3,12 |

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| | | | Marks | BL | CO | PO |
|---|---|---|---|---|---|---|
| IV. | | Describe the working of a 4-bit look ahead carry adder. Clearly show the derivation of the equations and implementation of hardware. | 10 | L2 | 1,2,3 | 1 |
| | | **OR** | | | | |
| V. | (a) | Implement the following function using 8×1 MUX $F = \sum m\,(0, 1, 2, 3, 4, 10, 11, 14, 15)$ | 5 | L3 | 1,2,3 | 1,2,3 |
| | (b) | Realize JK flipflop using SR flipflop. | 5 | L2 | 1,2,3 | |
| VI. | (a) | Design and setup a MOD-6 ripple up-counter using positive triggered JK flipflop. Show the timing diagram of the counter showing the glitches in the diagram. | 5 | L6 | 1,2,3 | 1,2,3 |
| | (b) | Design and setup a circuit to generate the following sequence by indirect method. 1 1 0 1 1 1 0 | 5 | L6 | 1,2,3 | 1,2,3 |
| | | **OR** | | | | |
| VII. | | Design and setup a synchronous sequential counter using JK flipflop and avoid lock-out condition for the sequence. $4 \rightarrow 6 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 4 \ldots$ | 10 | L6 | 1,2,3 | 1,2,3 |
| VIII. | (a) | Implement the following function using PLA $X(a, b, c, d) = \sum m(0, 2, 6, 7, 8, 9, 12, 13, 14)$ $Y(a, b, c, d) = \sum m(2, 3, 8, 9, 10, 12, 13)$ $Z(a, b, c, d) = \sum m(1, 3, 4, 6, 9, 12, 14)$. | 5 | L3 | 1,2,3 | 1,2,3,12 |
| | (b) | Implement a full adder using PROM. | 5 | L3 | 1,2,3 | 1,2,3,12 |
| | | **OR** | | | | |
| IX. | (a) | Draw and explain the working of a two input TTL NAND totem-pole configuration. | 5 | L2 | 1,2,3 | 1,12 |
| | (b) | Four 7-bit hamming codes are transmitted serially through a noisy channel. Detect and correct errors, if any, in the even parity hamming code. 1001001011100111101100011011 | 5 | L3 | 1.2 | 1,2,12 |

Bloom's Taxonomy Levels

L2 - 23.33%, L3 - 42.22% L6 - 34.45%.

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